

Set-up and registration

International Marketing Analytics



Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at

[Office hours: book online \(Google Calendar\)](#)

Today: from zero to a working set-up

Accounts

Tools

Repo

Python

Check

AI

What you need



- Your laptop and charger
- Windows 10/11 or a recent macOS
- About 10 GB free disk space
- Admin rights to install software

Have at hand



- Your WU e-mail and its password
- Your phone, for two-factor sign-in
- Your student ID or enrolment confirmation

How we work



- One step at a time, all together
- Check your neighbour's screen
- Stuck? Raise your hand, keep going with the next step



Goal: by the end of today the check script prints “ready” on every laptop.

Register your accounts



Part 1 of 4 · GitHub, GitHub Education,
DataCamp

Four registrations

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GitHub

Stores code and its history; hosts the course and group repositories

github.com/signup



GitHub Education

Free student benefits, including GitHub Copilot

education.github.com



DataCamp

Online Python courses; four certificates count 20 %

[Invitation link on Canvas](#)



Canvas survey

Tell us your GitHub username so we can give you access

[Course page on Canvas](#)



Same e-mail everywhere: your WU student address unlocks the free plans.

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1 Open github.com/signup

2 **WU e-mail:** sign up with your student address

3 **Username:** short and professional, e.g. anna-berger

4 **Verify** the code GitHub sends to your inbox

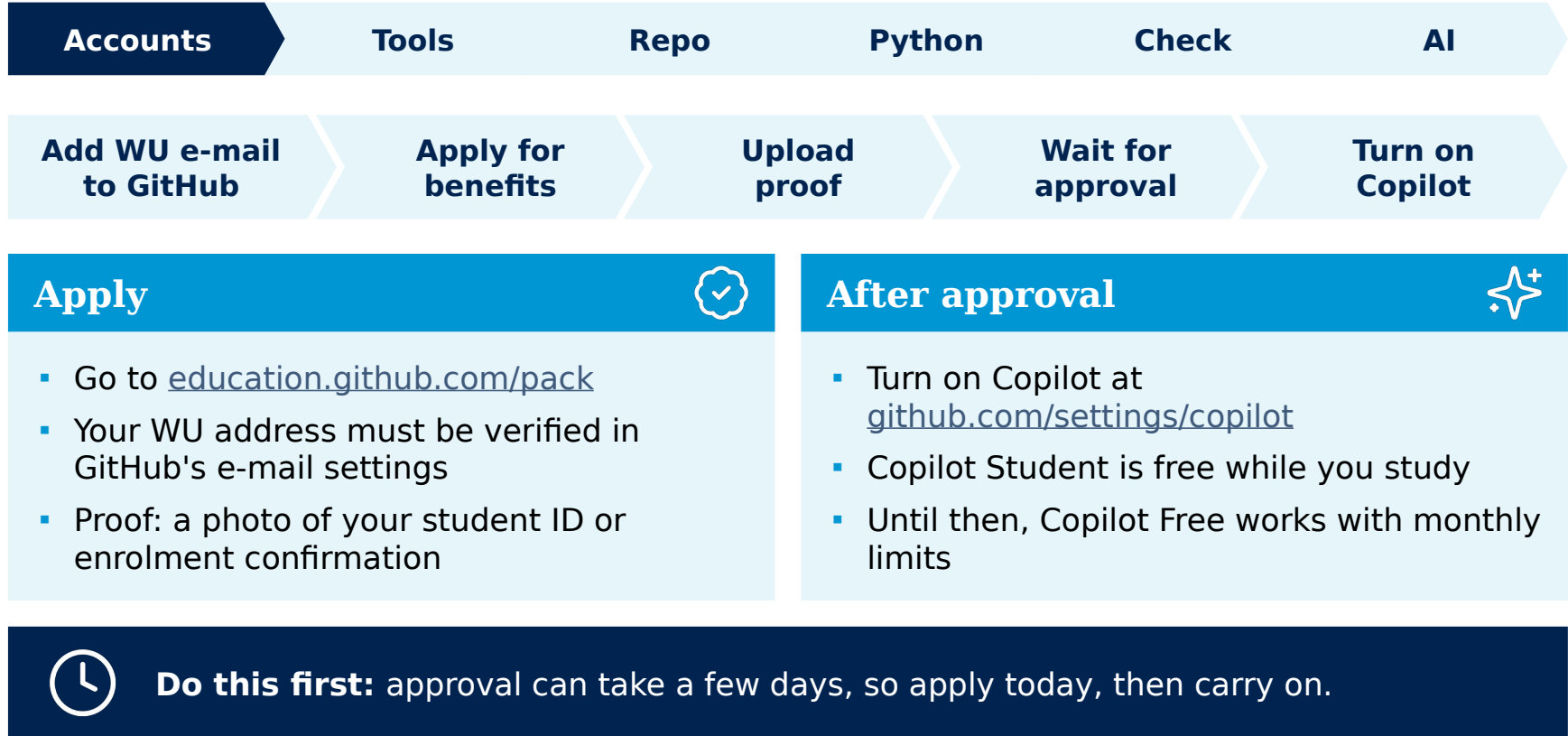
5 **Two-factor sign-in:** GitHub Mobile or an authenticator app

Why it matters



- Your username appears on every commit you make
- A clean GitHub profile is part of your CV
- You need it for the group repository of the case study
- The free plan is all you need

GitHub Education and Copilot



Accounts

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1 **Open** the invitation link on Canvas

2 **Sign up** with your WU e-mail, or log in

3 **Join** the classroom International Marketing Analytics

4 **Find** the four assigned courses under Assignments

5 **Upload** each certificate to Canvas by its deadline

6

months free premium access

4

courses with certificates

20 %

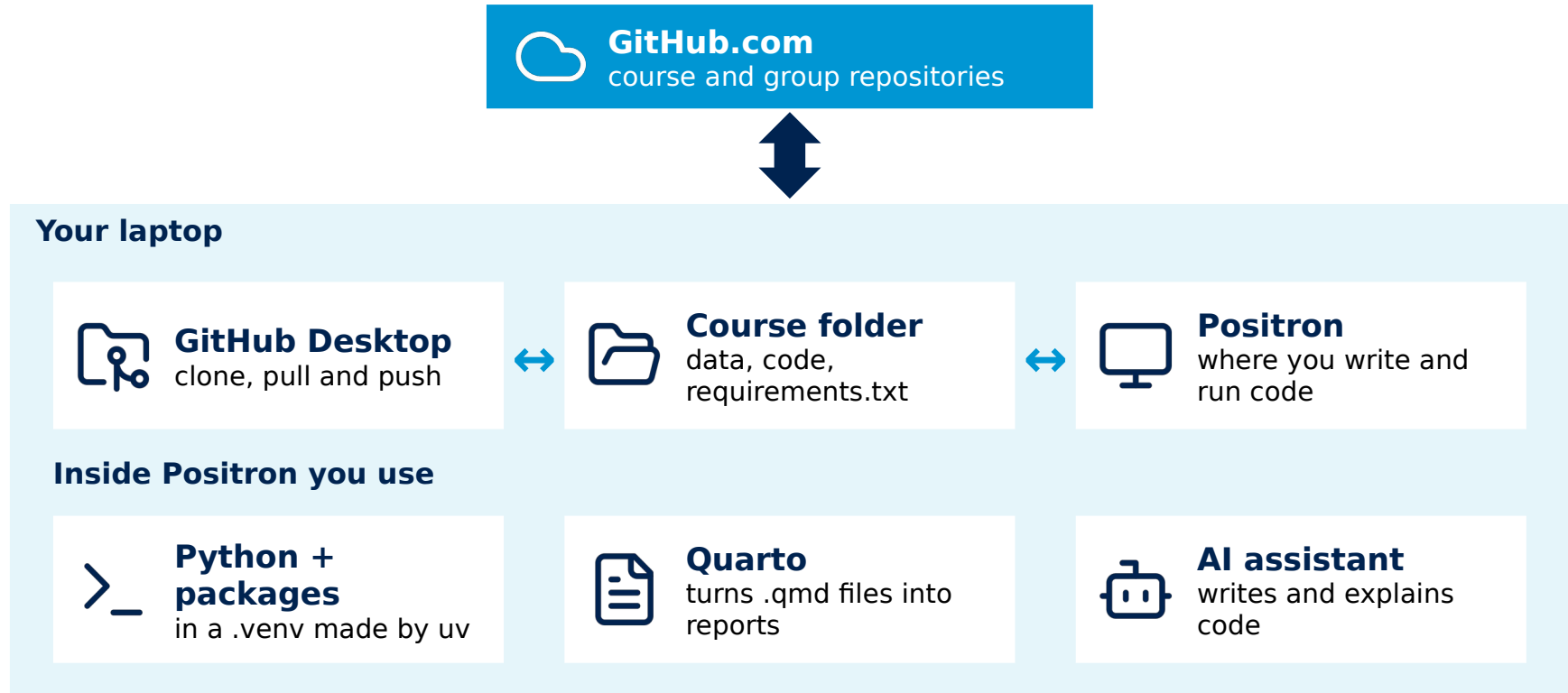
of the final grade

Install the tools



Part 2 of 4 · uv, Positron, Quarto, GitHub
Desktop, the course repository

How the pieces fit together



Step 2: install uv, which installs Python

Accounts

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Windows: PowerShell (one line)

```
powershell -ExecutionPolicy ByPass -c "irm  
https://astral.sh/uv/install.ps1 | iex"
```

macOS: Terminal

```
curl -Lsf https://astral.sh/uv/install.sh | sh
```

Then, in a new terminal window: check

```
uv --version
```

What uv does



- Installs Python for you
- Creates a private environment per project
- Installs packages in seconds
- Replaces pip, venv and pyenv

Windows: type the command on one line; the line break above is only for the slide.

Step 2: install Positron, Quarto and GitHub Desktop

Accounts

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Positron



- The editor for code, data and charts
- Free, from Posit
- macOS: drag it to Applications

positron.posit.co/download

Quarto



- Turns text and code into reports and slides
- Check: `quarto --version`

quarto.org/docs/get-started

GitHub Desktop



- Clone, pull and push with clicks
- Installs Git for you
- Sign in with your GitHub account

desktop.github.com



Restart Positron after installing Quarto, so it finds it.

Step 3: get the course repository

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Copy the URL
from Canvas

GitHub Desktop:
Clone repository

Pick a folder
without spaces

Positron:
Open Folder

Good folder names

C:\Users\anna\wu\ima
~/wu/ima

Avoid spaces, umlauts and cloud folders such as OneDrive or iCloud: they break paths and syncing.

Clone: your own copy of the repository, linked to GitHub

Pull: get the latest materials; do it before every session

Commit and push: save and upload your changes (group repository)



In Positron: File → Open Folder and choose the cloned folder.

Step 4: create the Python environment

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Positron: Terminal → New Terminal, then

```
uv venv --python 3.12
uv pip install -r requirements.txt
```

1 Creates .venv: a private Python just for this folder

2 Installs the packages and versions in requirements.txt

3 Select it: click the interpreter at the top right

The interpreter selector

Python 3.12 (.venv) ▼

- Shows which Python runs your code
- Pick the one that ends in .venv
- Once per folder; Positron remembers it

First time: a few minutes, because PyMC is large. After that you only pull and open the folder.

Step 5: run the check

Accounts

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Repo

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Check

AI

In the Positron terminal

```
uv run --no-project python setup/check_setup.py
```

Example output (shortened)

```
Packages
[ok] polars          1.44.2    data wrangling
[ok] plotnine        0.15.8    charts
[ok] great_tables    1.0.0     tables
[ok] statsmodels      0.15.0    regression
...
ready
```

Reading it



- [ok]: in place
- [!!!]: the line says what to fix
- Ends with “ready”: done



No terminal? Open setup/check_setup.py in Positron and press Run.

Step 6: connect your AI assistant

Accounts

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Check

AI

- 1 **Accounts menu** in Positron (bottom left):
sign in with GitHub
- 2 **Copilot** now suggests code as you type
- 3 **Posit Assistant:** open the chat, choose
GitHub Copilot

Options

Copilot Student: free, after GitHub
Education approval

Copilot Free: free, with monthly limits

Posit AI Pass: about USD 20 a month

Anthropic API key: pay as you go



Our rule: the AI writes, you check. Run the code, read the output and keep a short prompt log for each submission.

How Positron works



Part 3 of 4 · panes, running code, Quarto,
shortcuts

ima — Positron

Python 3.12 (.venv) ▾

- 1 Explorer**
the files of the
course folder

```
data/  
sessions/  
setup/  
requirements.txt
```

- 2 Editor**
write .py and .qmd files

```
sales = pl.read_csv(...)  
sales.head()
```

- 3 Console and Terminal**
code runs here, results print

```
>>> sales.height  
936
```

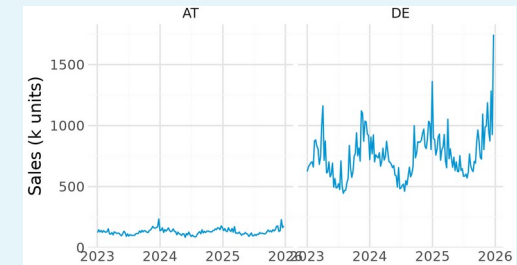
- 4 Variables**
what is in memory; click a
table to explore it

sales 936 × 18

summary 6 × 3

model OLS results

- 5 Plots**
charts appear here



From code to result

**Write code
in the editor**

**Ctrl/Cmd
+ Enter**

**It runs in
the console**

**Objects in
Variables**

**Charts in
Plots**

Run code



- Ctrl/Cmd + Enter: the line or selection
- Ctrl/Cmd + Shift + Enter: the current cell of a .qmd file
- Run button (top right): the whole file

Look at results



- Variables: every object, its type and size
- Data Explorer: sort, filter and summarise a table without code
- Plots: earlier charts, zoom, save as PNG



Console or terminal? The console speaks Python. The terminal speaks to your computer: uv, git and quarto commands go there.

Quarto documents in Positron

report.qmd

```
---  
title: "Sales by country"  
format: html  
---  
  
## Revenue in 2025  
  
``{python}  
import polars as pl  
sales = pl.read_csv("data/mmm/alphaglow_weekly.csv")  
``  
  
Germany has the highest revenue.
```



Text in Markdown: headings, bold, lists



Code cells run one by one, like a notebook



Preview: Ctrl/Cmd + Shift + K shows the report next to your code



Render to HTML, PDF, Word or slides



One file, rerun any time: the labs and your case study report are Quarto documents.

Shortcuts worth knowing

Action	Windows	macOS
Run line or selection	Ctrl + Enter	Cmd + Enter
Run current cell (.qmd)	Ctrl + Shift + Enter	Cmd + Shift + Enter
Preview a Quarto document	Ctrl + Shift + K	Cmd + Shift + K
Find any command	Ctrl + Shift + P	Cmd + Shift + P
Open a file by name	Ctrl + P	Cmd + P
Comment or uncomment lines	Ctrl + /	Cmd + /
New terminal	Ctrl + Shift + `	Ctrl + Shift + `
Accept an AI suggestion	Tab	Tab

Copilot: ghost text



- Grey suggestions appear as you type
- Tab accepts, Esc rejects
- Best for short, routine lines

Posit Assistant: chat



- Sees your data, variables and plots
- Writes, explains and fixes code
- You approve each step it runs

Good prompts

```
"Sum revenue by country for 2025 with polars and sort it."  
"Explain this error in two sentences, then fix it."
```



Privacy: prompts, variable names and types go to the model provider; rows of data only if you ask. Never paste personal data.

Python packages for this course



Part 4 of 4 · polars, plotnine, Great Tables,
statsmodels

Packages: install once, import every time

requirements.txt
lists packages

uv pip install
once per laptop

import
in every file

Top of every script

```
import polars as pl
import statsmodels.formula.api as smf
from plotnine import ggplot, aes, geom_line
from great_tables import GT
```

Conventions

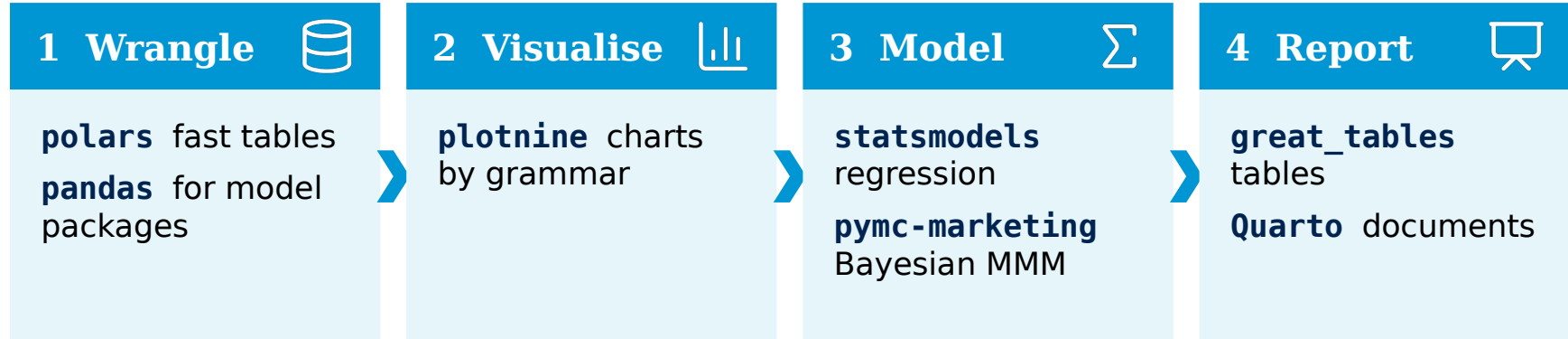


- pl and smf are the usual short names
- Imports go at the top
- Import only what you use



ModuleNotFoundError? The wrong interpreter is selected (pick .venv) or the package is missing (run `uv pip install -r requirements.txt`).

Your toolkit, from data to report



22

packages pinned in requirements.txt

1

command installs them all

0

euros: all open source



Online you will also meet pandas and matplotlib: older but common; the ideas carry over.

Revenue and TV spend per country, 2025

```
import polars as pl

sales = pl.read_csv("data/mmm/alpenglow_weekly.csv",
                    try_parse_dates=True)

summary = (
    sales
    .filter(pl.col("week").dt.year() == 2025)
    .group_by("country")
    .agg(pl.col("revenue_eur_k").sum().alias("revenue"),
         pl.col("spend_tv_k").sum().alias("tv"))
    .sort("revenue", descending=True)
)
```

DataFrame: a table with typed columns

Expressions: `pl.col("tv").sum()`
says what to compute

Chaining: one step per line, read top to bottom

Fast: multi-threaded, written in Rust

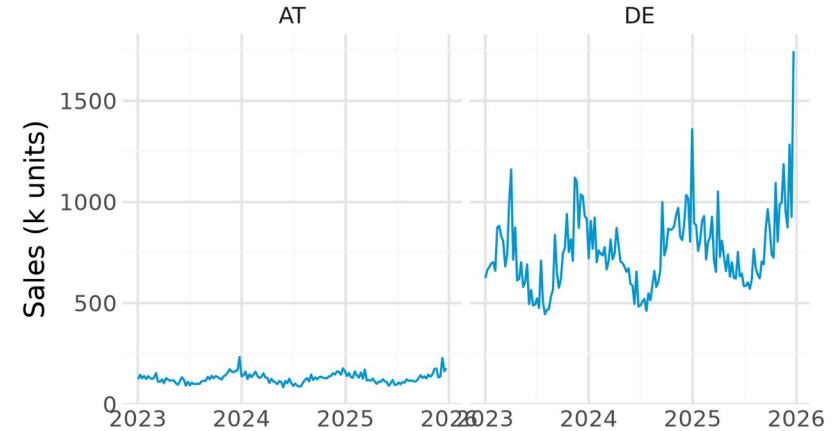


Result: six rows, one per country. Germany leads with about EUR 131 million revenue in 2025. Need pandas? Add `.to_pandas()`.

plotnine: charts with the grammar of graphics

Weekly sales, Austria and Germany

```
(ggplot(sales.filter(
  pl.col("country").is_in(["AT", "DE"])),
  aes(x="week", y="sales_units_k"))
+ geom_line(color="#0096D3")
+ facet_wrap("country")
+ scale_x_date(date_breaks="1 year",
  date_labels="%Y")
+ labs(x="", y="Sales (k units)")
+ theme_minimal())
```



The output of the code on the left

ggplot()
the data

aes()
columns to x, y

geom_
the marks

facet_
one panel per
group

theme_
the look

Data + mapping + marks + panels + theme: each + adds one layer.

Great Tables: tables for reports

The polars summary as a table

```
from great_tables import GT

(GT(summary)
 .tab_header(title="Revenue and TV spend, 2025")
 .fmt_number(columns=["revenue", "tv"], decimals=0)
 .cols_label(country="Country",
              revenue="Revenue (k EUR)",
              tv="TV spend (k EUR)"))
```

Header and labels: readable names instead of column codes

Number formats: thousands separators, currency, percent

Works with polars and pandas; renders in Quarto reports

Revenue and TV spend, 2025		
Country	Revenue (k EUR)	TV spend (k EUR)
DE	130,650	5,434
FR	66,306	2,363
IT	47,950	2,478
NL	33,234	812
PL	23,112	616
AT	21,984	695

Weekly sales in Austria, 2023 to 2025

```
import statsmodels.formula.api as smf

at = sales.filter(pl.col("country") == "AT").to_pandas()
model = smf.ols(
    "sales_units_k ~ price_eur + spend_tv_k"
    " + spend_paid_search_k",
    data=at).fit()
print(model.summary())
```

Coefficient	Estimate
Intercept	250.63
price_eur	-48.31
spend_tv_k	0.41
spend_paid_search_k	4.03
R ² (n = 156)	0.43



Reading it: EUR 1,000 more paid search in a week goes with about 4,000 more units sold, holding price and TV constant.

Later in the course: scikit-learn for logistic regression and pymc-marketing for Bayesian MMM.

When something goes wrong

Problem	Fix
uv: command not found	Close and reopen the terminal, or restart Positron
PowerShell: running scripts is disabled	Use the exact uv command; it includes -ExecutionPolicy Bypass
ModuleNotFoundError	Select the .venv interpreter; run uv pip install -r requirements.txt again
Quarto preview does not start	Restart Positron; run quarto check in the terminal
Clone fails in GitHub Desktop	Sign in again; check the URL; is your username in the Canvas survey?
Copilot is not offered	GitHub Education still pending: use Copilot Free meanwhile

Checklist: before you leave today

Accounts



- ☐ GitHub account with two-factor sign-in
- ☐ GitHub Education application sent
- ☐ DataCamp classroom joined
- ☐ GitHub username in the Canvas survey

Laptop



- ☐ uv, Positron, Quarto and GitHub Desktop installed
- ☐ Course repository cloned and open
- ☐ .venv created and selected
- ☐ Check script prints “ready”
- ☐ AI assistant signed in



Not ready yet? Post the failing step in the Q&A forum on Canvas before Session 2.



VIENNA UNIVERSITY OF
ECONOMICS AND BUSINESS

**INSTITUTE FOR INTERNATIONAL
BUSINESS**

Welthandelsplatz 1, 1020 Vienna,
Austria

DR ARNE FLOH

Senior Lecturer in Marketing

F +43-31336-6367

arne.floh@wu.ac.at

www.wu.ac.at/en/welthandel/arne-floh

[Office hours: book online](#)